



**BINAY PowerLED Inspection Hand Lamps utilize a light source comprised of multiple high power white LEDs**

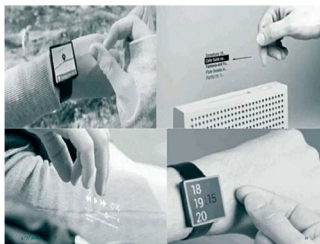
- Very low power consumption of 10 watts or less
- Long life (over 50,000 burning hours), covered under 3-year warranty
- Unbreakable shock-proof and vibration-resistant construction; cool light output with no heat radiation
- Glare-free light output, focusing only on the work area



**Binay Opto Electronics Pvt. Ltd.**

27, PANDITA TERRACE, Calcutta 700 029, India  
Telephone: +91 (33) 4006 9875, 2475 0392, 2475 0030  
email: info@binayled.com, http://www.binayled.com

**POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983**



Project Soli sensor (Image courtesy: <https://image.slidesharecdn.com/>)

## MEMS combo sensors

MEMS sensors are the heart of smartphones, enabling smart functions. Niranjan G., general manager (business development), Rohm Semiconductor Singapore, says, "As MEMS sensors consume less power, these are making their way successfully into mobile devices."

MEMS sensors were used for microphone applications too, but it was difficult to give an input that yielded noiseless output. Industrial-grade MEMS sensors have now reached around 70dB signal-to-noise ratio (SNR). That means voice commands can be captured from a long distance and traced with minimal losses and interference. The higher the SNR ratio, the lower is the noise produced.

This year, the market for MEMS sensors is said to touch one billion dollars. Not just in microphones, MEMS sensors are going to play a major role in the advancement of automobiles too. For instance, the ADAS system in cars combines MEMS sensors for different functionalities such as gyroscope, accelerometer and speedometer in one place. Thus, smart use of combo MEMS sensors has simplified large boards with different applications and cut down manufacturing costs.

Last year, Tesla announced the use of sensors for its complete electric car solution. Similarly, the market is said to beam up with Lidar sensor technology, which is specially developed for image processing by cameras in automobiles.

Lenovo, Motorola and Xiaomi are among the key customers of combo MEMS sensors.

## Go wireless with Wi-Sun

Smart cities and industries will use the IoT, which will be primarily wireless based. Sensors deployed for various applications will sum up to form a wireless system. So wireless sensors are enablers for the IoT and the Industrial IoT (IIoT).

# IT'S TIME TO PROMOTE DESIGN IN INDIA

<http://designindia.electronicshor.com>



**Design IN India**

Section launched on Electronicsforu.com to showcase design houses and innovators of India who are creating innovative electronic hardware products. If your firm has also launched or designed an innovative product for OEMs, share your details with us at [editsec@efy.in](mailto:editsec@efy.in).

**AN EFGROUP  
PUBLICATION**